SC12 - Revision H purchasing BOM

140 fitted parts: 130 factory SMT + 10 user-fitted through-hole parts. Three loose caps extra. Parts estimate: $25.61/board at 10 boards before SMT attrition; PCB, assembly, freight and tax extra. JLC Standard PCBA candidate: MCU stock and C1 reflow unresolved; simulation pending approval. See docs/JLC_SOURCING.md.

Qty	MPN / supplier part	References	Value / package	Extended USD first tier
1	0451003.MRL / C13982	F1	3A FAST / Fuse_Littelfuse-NANO2-451_453	0.2555
4	0805W8F1001T5E / C17513	R4, R110, R210, R310	1k / R_0805_2012Metric	0.0168
12	0805W8F1002T5E / C17414	R3, R21, R22, R40, R41, R105, R205, R305, R52, R71, R75, R31	10k / R_0805_2012Metric	0.0468
8	0805W8F1003T5E / C149504	R50, R111, R112, R211, R212, R311, R312, R70	100k / R_0805_2012Metric	0.0472
1	0805W8F1152T5E / C17418	R61	11.5k / R_0805_2012Metric	0.0041
3	0805W8F2203T5E / C17556	R101, R201, R301	220k / R_0805_2012Metric	0.015
1	0805W8F3300T5E / C17630	R30	330 / R_0805_2012Metric	0.0054
1	0805W8F5231T5E / C17739	R51	5.23k / R_0805_2012Metric	0.0023
1	0805W8F5902T5E / C17763	R10	59k / R_0805_2012Metric	0.0023
1	1206W4F1200T5E / C17909	R42	120 / R_1206_3216Metric	0.0074
3	1824420000 / 281-2020-ND	J1, J2, J3	2P PUSH 3.5mm / Weidmuller_LSF_90_2P	4.89
3	1824430000 / 281-2021-ND	J4, J5, J6	3P PUSH 3.5mm / Weidmuller_LSF_90_3P	6.63
1	ABM3B-8.000MHZ-B2-T / C83312	Y1	8MHz 18pF / Abracon_ABM3B_5.0x3.2mm	0.589
3	AO3401A / C15127	Q101, Q201, Q301	AO3401A / SOT-23	0.2829
1	AO4485 / C51499	Q1	AO4485 / SOIC-8_3.9x4.9mm_P1.27mm	0.295
1	AP2112K-3.3TRG1 / C51118	U2	AP2112K-3.3 / SOT-23-5	0.1712
2	B240A-13-F / C92506	D4, D5	B240A / D_SMA	0.1902
3	BAT46W-E3-18 / C5557780	D102, D202, D302	BAT46W / D_SOD-123	0.3927
5	BSS138BK,215 / C282529	Q102, Q202, Q302, Q2, Q3	BSS138BK / SOT-23	0.3345
3	CC0805JRNPO9BN102 / C62774	C101, C201, C301	1nF / C_0805_2012Metric	0.0411
2	CC0805JRNPO9BN300 / C107114	C25, C26	30pF / C_0805_2012Metric	0.0146
1	CC0805KRX7R9BB333 / C107150	C69	33nF / C_0805_2012Metric	0.012
2	CL21B103KBANNNC / C1710	C72, C73	10nF / C_0805_2012Metric	0.026
14	CL21B104KCFNNNE / C28233	C3, C6, C11, C20, C21, C22, C27, C40, C41, C50, C51, C66, C74, C75	100nF / C_0805_2012Metric	0.5698

Qty	MPN / supplier part	References	Value / package	Extended USD first tier
5	CL21B105KBFNNNE / C28323	C7, C23, C61, C65, C71	1uF / C_0805_2012Metric	0.2165
1	CL21B473KBCNNNC / C53134	C70	47nF / C_0805_2012Metric	0.0173
2	CL21B475KAFNNNE / C98195	C8, C24	4.7uF / C_0805_2012Metric	0.1214
8	CL31A226KAHNNNE / C12891	C4, C5, C13, C12, C60, C62, C63, C64	22uF / C_1206_3216Metric	1.3544
1	CL32B106KBJNNWE / C3844168	C2	10uF / C_1210_3225Metric	0.7143
1	EEE-FK1H470P / C278400	C1	47uF / CP_Elec_8x6.5	0.2942
1	ESD2CAN24DBZRQ1 / C5736151	D20	ESD2CAN24-Q1 / SOT-23	0.3486
1	LM339LVDR / C22427665	U7	LM339LV / SOIC-14_3.9x8.7mm_P1.27mm	0.5858
1	LMR51420XDDCR / C5246146	U1	LMR51420XDDCR / TSOT-23-6	1.0218
3	M22-2010305 / 952-2274-ND	JP101, JP201, JP301	5V / 12V / PinHeader_1x03_P2.00mm_Vertical	0.66
1	MMSZ12T1G / C72599	D3	12V ZENER / D_SOD-123	0.0723
1	PM254V-12-10P-H85 / C492399	J8	2x5 SERVICE SOCKET 2.54mm / Service_2x05_P2.54mm	0.1158
1	RC0805FR-0713KL / C137592	R63	13k / R_0805_2012Metric	0.0055
3	RC0805FR-0720KL / C114566	R103, R203, R303	20k / R_0805_2012Metric	0.0237
1	RC0805FR-072K61L / C273965	R62	2.61k / R_0805_2012Metric	0.0075
3	RC0805FR-072M2L / C137552	R104, R204, R304	2.2M / R_0805_2012Metric	0.0219
1	RC0805JR-070RL / C96345	R20	0 / R_0805_2012Metric	0.0066
3	RC1206FR-0710KL / C132649	R109, R209, R309	10k / R_1206_3216Metric	0.0333
6	RT0805BRD07100KL / C122537	R60, R65, R66, R68, R72, R73	100k / R_0805_2012Metric	0.2622
1	RT0805BRD07102KL / C865126	R1	102k / R_0805_2012Metric	0.0471
1	RT0805BRD0712K1L / C865169	R69	12.1k / R_0805_2012Metric	0.0533
1	RT0805BRD0713K7L / C865183	R2	13.7k / R_0805_2012Metric	0.0467
2	RT0805BRD0720KL / C469660	R67, R74	20k / R_0805_2012Metric	0.1484
1	SK6805-EC15 / C2890035	D10	SK6805-EC15 / LED_SK6805_EC15	0.0995
1	SMAJ13A-13-F / C110519	D6	SMAJ13A / D_SMA	0.124
3	SMAJ30A / C148230	D101, D201, D301	SMAJ30A / D_SMA	0.4359
1	SMBJ22A-E3/52 / C3698580	D2	SMBJ22A / D_SMB	0.467
1	SN74AHCT1G125DBVR / C7484	U10	SN74AHCT1G125 / SOT-23-5	0.1001
1	SRP5030TA-3R3M / C2042118	L2	3.3uH / L_Bourns_SRP5030T	2.3951
1	SRP7028A-100M / C2687402	L1	10uH / L_Bourns_SRP7028A_7.3x6.6mm	0.9356

Qty	MPN / supplier part	References	Value / package	Extended USD first tier
1	STM32F042K6T6 / C69216	U5	STM32F042K6T6 / LQFP-32_7x7mm_P0.8mm	2.9157
1	TCAN3413DR / C22433320	U6	TCAN3413DR / SOIC-8_3.9x4.9mm_P1.27mm	1.0153
1	TPS2553DBVR-1 / C111738	U4	TPS2553DBVR-1 / SOT-23-6	0.2911
1	TPS259620DDAR / C2155773	U9	TPS259620DDAR / TPS2596_DDA8	0.8152
1	TPS61085DGKR / C113659	U8	TPS61085DGKR / VSSOP-8_3x3mm_P0.65mm	0.6004